Example: (Classical Gram-Schmidt Orthogonalization)

① Find $\vec{q}_1, \vec{q}_2, \vec{q}_3$ (orthonormal)

from $\vec{a}_1, \vec{a}_2, \vec{a}_3$ (columns of A)

② Also find the coefficients of r_{ij} in the R matrix.

$$\boxed{A = QR}$$

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}$$

Sol^n: $\boxed{\text{Find } \vec{q}_1 \text{ using } \vec{a}_1}$

Step 1: To find $\vec{q}_1$, we start with $\vec{a}_1$ & make it orthonormal to previous one.

There is no previous a_0, so the direction of $\vec{a}_1$ is fine, Also length should be 1.

$$\Rightarrow q_1 = \frac{a_1}{\|a_1\|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Step 2: Find $\vec{q}_2$ using $\vec{a}_1, \vec{a}_2$ & $\vec{q}_1$

- q_2 should be orthogonal to $\vec{q}_1$
- q_2 should have length 1.

$$\Rightarrow \boxed{\vec{q}_2' = \vec{a}_2 - (\vec{a}_2^* q_1)\vec{q}_1^*} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - 2\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

Start with $\vec{a}_2$

make $\vec{a}_2$ orthogonal to $\vec{q}_1$

(further & normalize it)

$$\Rightarrow q_2' = \frac{q_2'}{3} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\|q_3'\|$.

Step 3: (do it yourself!)

- Construct $\vec{q}_3$ using a_3 and make $\vec{a}_3$ orthogonal to $\vec{q}_1$ & $\vec{q}_2$
- q_3 should have length 1.

Answer: $q_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

We have found Q using Gram-Schmidt orthogonalization

So what is R?

Let's see:

Notice eq ①: $\vec{q}_1 = \frac{\vec{a}_1}{\|a_1\|} \Rightarrow \vec{a}_1 = \overset{r_{11}}{\boxed{1}}\vec{q}_1$

eq ②: $3\vec{q}_2 = a_2 - 2q_1$

$\Rightarrow a_2 = \underset{r_{12}}{\boxed{2}}\vec{q}_1 + \underset{r_{22}}{\boxed{3}}\vec{q}_2$

eq ③: DO IT YOURSELF.

Find r_{13}, r_{23}, r_{33}.

$$\Rightarrow R = \begin{bmatrix} 1 & 2 & ? \\ 0 & 3 & ? \\ 0 & 0 & ? \end{bmatrix}$$

(So we've have $QR = A$)

(You can verify this!)